

Assignment of Chapter 4

A. Fill-in-Blank:

1. Each flip-flop consists of _____ (how many) steady states; each flip-flop can save _____ (how many) bits of binary number.
2. the next state equation of the T flip-flop is _____.
3. A basic flip-flop device made of NAND gates cannot have the _____ as its input.
4. _____, _____ and _____ these three clock-controlled flip-flops have no constraints on input.
5. With the clock pulse on, in order to enable a J-K flip-flop to have $Q^{n+1} = \bar{Q}$, thus, the input signal must be J=_____, K=_____.
6. Once the clock pulse is on, a flip-flop's input can affect the next state of the flip-flop. This statement is _____. (true/false)
7. A combinational logic circuit must not have any memory devices. This statement is _____. (true/false)
8. 4 flip-flop devices can save up to 2 bits binary numbers. This statement is _____. (true/false)
9. A basic R-S flip-flop made of NOR gates cannot have RS=00, such inputs. This statement is _____. (true/false)
10. A clock based R-S flip-flop cannot have the situation that its both outputs are equivalent. This statement is _____. (true/false)

B. Short Answer:

1. What is the difference in logic function between a basic R-S flip-flop made of NAND gates and a basic R-S flip-flop made of NOR gates?

2. Fig 1 illustrate a D flip-flop diagram, and its input waveform is shown in Fig 2, please draw the output waveform of Q. (Assume the present state of Q is 0)

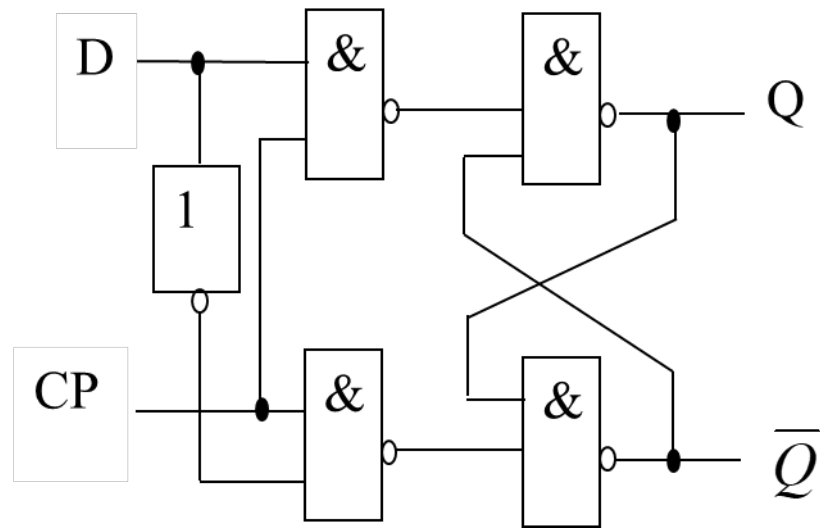


Fig 1

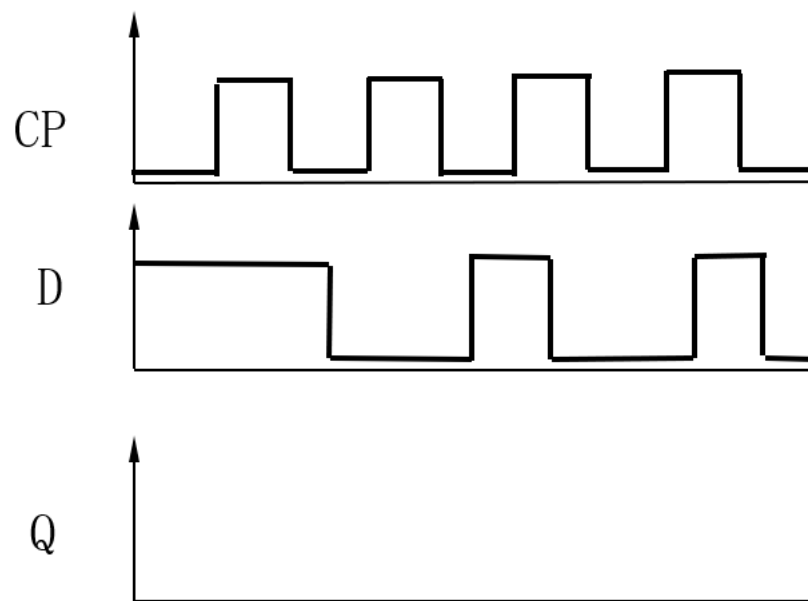


Fig 2